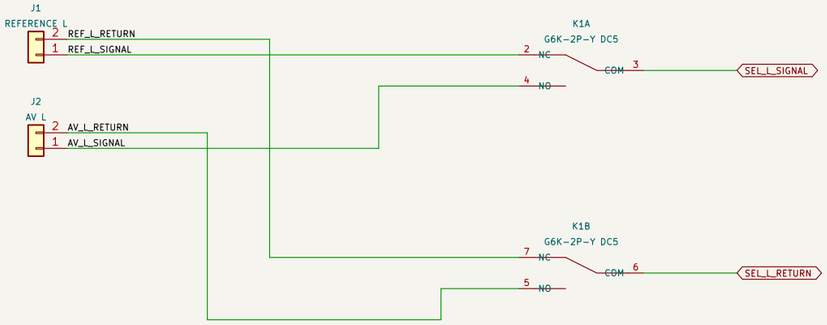
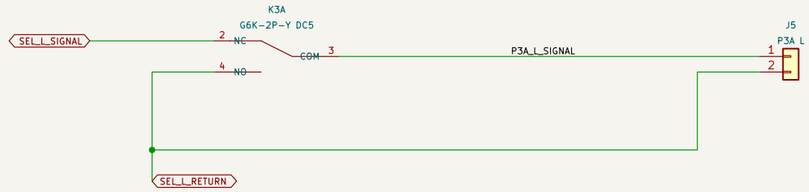
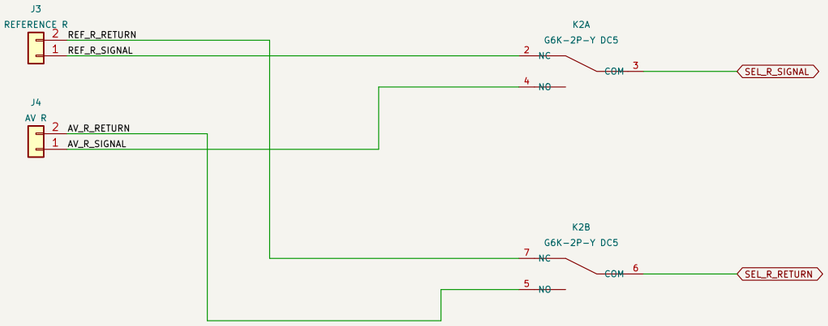
CONTACTS ONLY IN THE THROUGH SIGNAL PATH

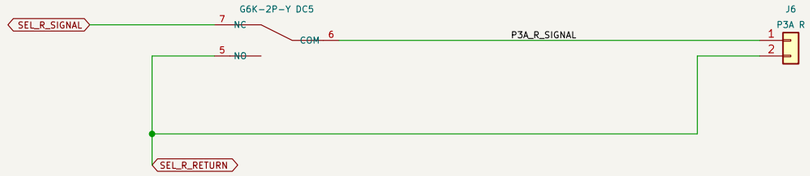
K1/K2 OFF = REFERENCE. ON = AV. K3 OFF = PASS. ON = output shorted to its own return.
All RCA shells must be insulated from the enclosure. Keep each signal with its corresponding return.





L channel: no common GND





R channel: no common GND

Sensing taps are on SEL_* before K3. No permanent connection exists to inactive inputs.
The P3A may join its own L/R returns externally: this board does not join them.
Unpowered pass-through is intentional: abrupt controller loss is not a guaranteed mute.

12V input and 5V buck supply

File: Power.kicad_sch

ATtiny412 and relay drivers

File: Control.kicad_sch

Isolated left detector

File: Detector_L.kicad_sch

Isolated right detector

File: Detector_R.kicad_sch

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

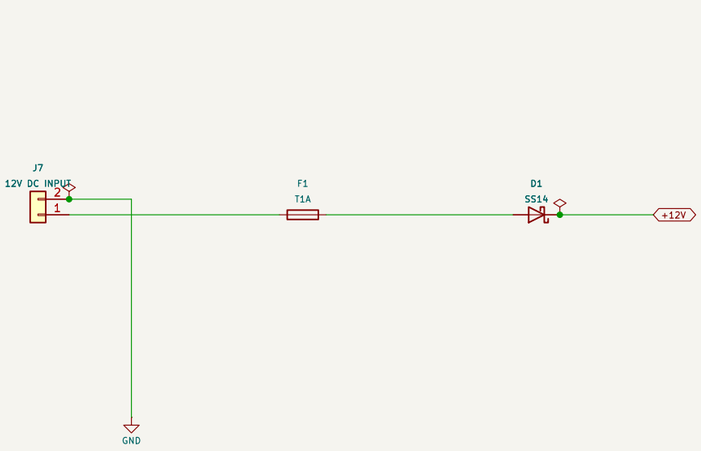
Sheet: /
File: InputSwitch.kicad_sch

Title: Audio source selector and output mute

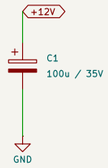
Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/5

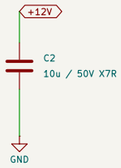
ISOLATED REGULATED 12V WALL ADAPTER: 12V +/−2%, >=1A

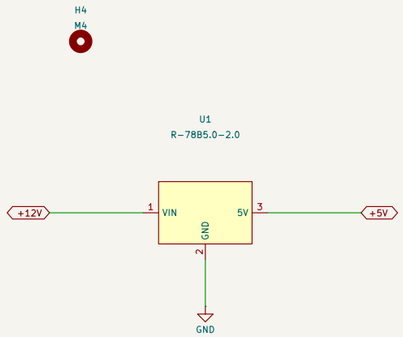
DC input only. G7L mains contacts and all 230V wiring are chassis−mounted off−board.
PE remains continuous and is never switched. Supply GND is the controller ground only.

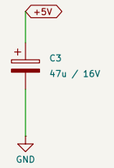


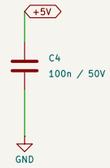


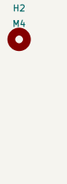


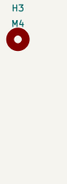












D1 reverse−polarity blocking; F1 limits fault current; D2 clamps short input transients.
Not a sustained−overvoltage disconnect. Use only the specified regulated adapter.
SS14 drop reduces the coil/buck input slightly below the adapter voltage.

5V supply: 2A capacity. Budget 1.1A including both SPU03 modules.
Detector preloads dissipate about 1.67W total at nominal voltage.
12V adapter >=1A; input fuse T1A. Verify startup and temperatures.
Ripple/EMI and fuse coordination remain bench/layout checks.







NO PCB MAINS. Do not update the old PCB from this schematic until layout work is authorised.

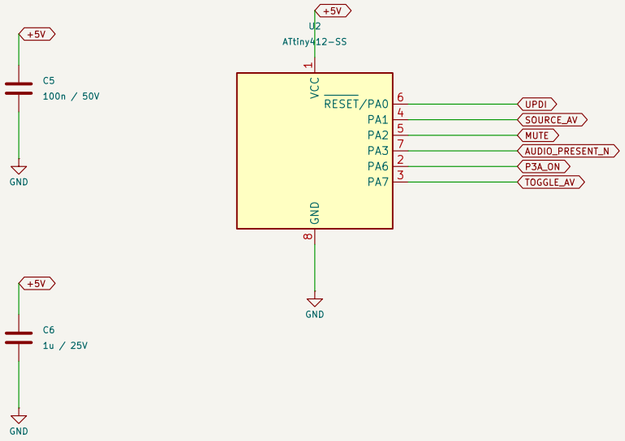
ULN2003A/G6K pickup margin requires bench validation.
Low−voltage controller only. No mains on this PCB.

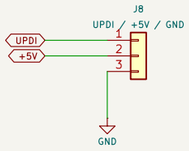
Sheet: /12V input and 5V buck supply/
File: Power.kicad_sch

Title: 12V input and 5V buck supply

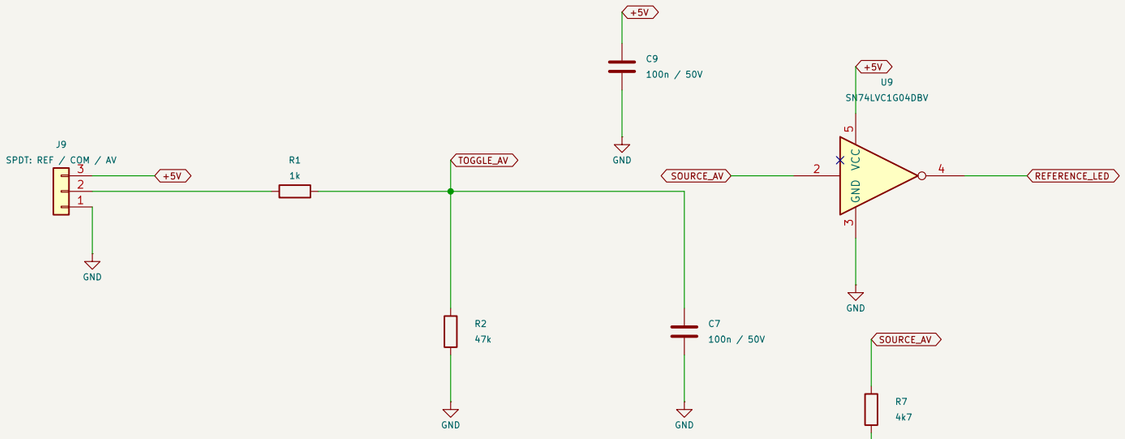
Size: A3	Date: 2026−10−01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/5

FIVE GPIOs; PA0 REMAINS UPDI



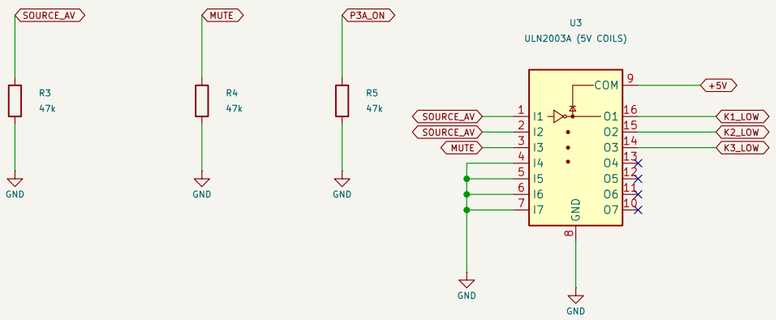


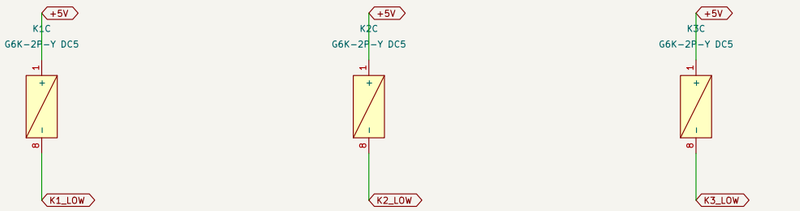
J8 pin 1 UPDI; pin 2 target-voltage reference; pin 3 GND.
Programmer must not back-power the board. BOD 4.2V + watchdog in firmware.



Open/disconnected switch -> REFERENCE.
100us contact-driven RC; 4.8ms with open switch.
Firmware debounce >=20ms.

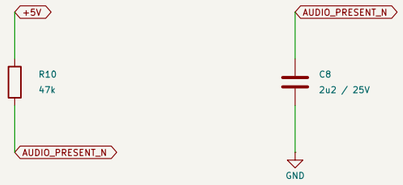
LEDs indicate source COMMAND.
They do not verify relay contacts.





Driver inputs tied low when MCU is reset.
Unused ULN inputs grounded; outputs genuinely unused.
No parallel output channels.

G7L-2A-T DC12: J10 pin 1 coil+; pin 2 low-side.





Optocoupler wired-OR: LOW = recent audio. 103ms nominal release RC.
Firmware must require 500ms persistence, then time 10min silence.
Blank detection at startup, source changes and after power-off.
Do not count a single edge/click as continuous audio.

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

Sheet: /ATtiny412 and relay drivers/
File: Control.kicad_sch

Title: ATtiny412 and relay drivers

Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/5

L CHANNEL — INDEPENDENT GALVANIC ISOLATION

Isolated return is tied ONLY to SEL_L_RETURN, after its source-selection relay.
No detector wire, resistor, capacitor or shield connects this return to control GND or the other channel.

U10
SPU03L-05
+5V
1
VIN+
VOUT+
7
ISO_L_5V
2
VIN-
VOUT-
5
SEL_L_RETURN
GND

+5V
C20
10u / 50V X7R
GND

ISO_L_5V
R20
30R / 3W / 1%
SEL_L_RETURN

ISO_L_5V
R21
100R
ISO_L_QUIET

ISO_L_QUIET
C21
47u / 25V
SEL_L_RETURN

ISO_L_QUIET
C22
100n / 50V
SEL_L_RETURN

ISO_L_QUIET
U5C
LM358N
8
V+
4
V-
SEL_L_RETURN

ISO_L_QUIET
R22
3k3
BIAS_L
R34
1k
SEL_L_RETURN

BIAS_L
C23
33u / 16V
SEL_L_RETURN

30R preload: ~167mA / 0.83W at 5V. Use 3W flameproof; keep clear of capacitors.
100R + 47u filters switching ripple. Bias ~1.1V; LM358 retained.
SPU03 INPUT IS 5V ONLY. Check isolated raw rails 4–6V and quiet rails >=3.6V.

ISO_L_5V
R32
390R
U6
VO617A-3
1
2
4
3
R33
1k
AUDIO_PRESENT_N
Q2
BC549
GND

BIAS_L
R29
2M2
BIAS_L
R27
10k
BIAS_L
R24
100k
SEL_L_SIGNAL
C24
100n FILM / 50V
R23
470k
SENSE_INPUT
U5A
LM358N
U5B
LM358N
DETECT_PULSE
R30
4k7
SEL_L_RETURN
R25
100k
R28
100k
SEL_L_RETURN
R31
100k
SEL_L_RETURN

CONTROL SIDE

Optocoupler collector is the only
logic interface. Both channels
wire-OR into one active-low input.

ISO_L_QUIET
D21
1N4148
SENSE_INPUT

SENSE_INPUT
D22
1N4148
SEL_L_RETURN

R26
1k
C25
33u / 16V
SEL_L_RETURN

Nominal midband input impedance ~570k. AC gain ~101 x 100/570 = 17.7.
Nominal trigger ~4mVrms at 1kHz; tuning/bench value, not a guaranteed noise threshold.
At 2Vrms the detector saturates intentionally; it is outside the through audio path.
470k limits loading when unpowered. Design input: up to 2Vrms sine (2.83V peak).

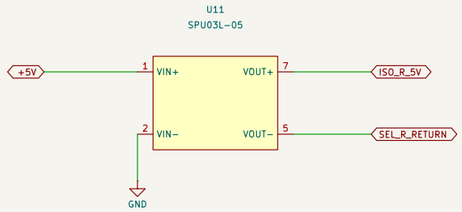
Branch clamps only;
never in the main audio path.

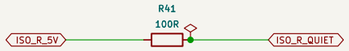
ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

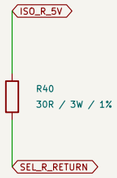
Sheet: /Isolated left detector/		
File: Detector_L.kicad_sch		
Title: Isolated left detector		
Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/5

R CHANNEL — INDEPENDENT GALVANIC ISOLATION

Isolated return is tied ONLY to SEL_R_RETURN, after its source-selection relay.
No detector wire, resistor, capacitor or shield connects this return to control GND or the other channel.

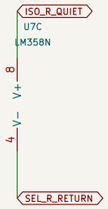






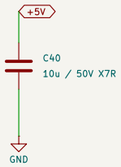




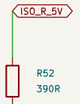


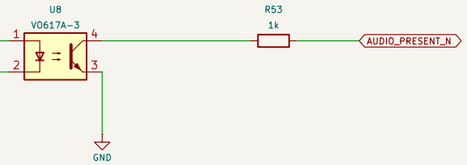




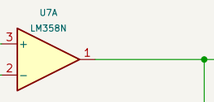


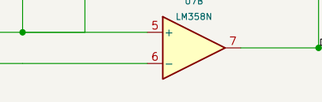
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100R + 47u filters switching ripple. Bias ~1.1V; LM358 retained.
SPU03 INPUT IS 5V ONLY. Check isolated raw rails 4–6V and quiet rails >=3.6V.

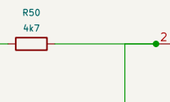




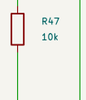






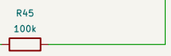


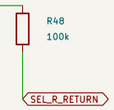






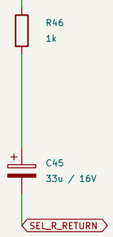












Nominal midband input impedance ~570k. AC gain ~101 x 100/570 = 17.7.
Nominal trigger ~4mVrms at 1kHz; tuning/bench value, not a guaranteed noise threshold.
At 2Vrms the detector saturates intentionally; it is outside the through audio path.
470k limits loading when unpowered. Design input: up to 2Vrms sine (2.83V peak).

CONTROL SIDE

Optocoupler collector is the only
logic interface. Both channels
wire-OR into one active-low input.

Branch clamps only;
never in the main audio path.

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

Sheet: /Isolated right detector/
File: Detector_R.kicad_sch

Title: Isolated right detector

Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/5